

Emissions Trading in the Climate Policy Assessment Tool: Design Options and Policy Interactions^{*}

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December 5, 2025

Abstract

Emissions Trading Systems (ETS) vary widely in design, particularly in whether they impose an absolute cap or an intensity-based limit, and in how permits are allocated. This paper introduces how different types of ETS and Tradable Performance Standards (TPS) are modeled in the World Bank–IMF Climate Policy Assessment Tool (CPAT), with emphasis on power sector modeling and policy interactions. We define key ETS types—mass-based cap-and-trade versus intensity-based (rate-based)—and allocation methods including auctioning, grandfathering, output-based allocation (OBA), and benchmarking. We show how each design affects emissions, carbon prices, and modeling needs, using simple equations where helpful. Permit benchmarks influence both stringency and output incentives. Case studies of the EU, China, India, and the US compare ETS/TPS instruments in the power sector, highlighting how intensity-based systems (e.g., China’s national ETS, India’s PAT scheme) differ from cap-based ones. A comparison table summarizes design features across the four. We then describe how these policy types are implemented in CPAT’s mitigation module. An extended appendix details CPAT’s “engineer” power model, which simulates electricity capacity, generation, and emissions.

Keywords: emissions trading; market-based regulations; abatement cost; renewable energy.

JEL Codes: Q41, Q48, Q54, Q58

^{*}The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

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1 Introduction

Market-based emissions trading has become a cornerstone of climate policy in many jurisdictions. An Emissions Trading System (ETS) caps total greenhouse gas emissions from covered sources and establishes a market for tradable emission permits (allowances). By putting a price on carbon, ETSs incentivize emitters to reduce emissions where it is cheapest to do so.

Over the past two decades, dozens of ETS programs have emerged worldwide, including the European Union Emissions Trading System (EU ETS), regional systems in North America (e.g. the Western Climate Initiative in California and the Regional Greenhouse Gas Initiative in the US Northeast), national schemes in East Asia (e.g. Korea, China), and pilot or upcoming systems in developing countries (e.g. a national carbon market in India building on its Perform, Achieve and Trade scheme) ([Commission, 2025](#); [CARB, 2022](#); [ICAP, 2022a](#); [PMR, 2021](#)). These systems vary widely in scope and design.

For instance, the EU ETS imposes an absolute cap on emissions and has moved toward predominantly auctioning allowances, whereas China’s national ETS uses an intensity-based “tradable performance standard” (TPS) with free allocation proportional to output ([ICAP, 2022a](#)). Design choices—such as whether an ETS is mass-based or intensity-based, and how allowances are allocated—have major implications for environmental effectiveness, carbon price levels, economic efficiency, and interactions with other policies.

A mass-based or “cap-and-trade” ETS sets a fixed emissions limit and lets the market determine the price needed to meet that cap ([Burtraw et al., 2018](#)). By contrast, a rate-based system imposes an emissions-per-unit-of-output limit; total emissions adjust with production, and the market price of credits ensures the average intensity target is met ([Fischer, 2006](#)). Allowance allocation—auctioning vs. free allocation based on formulas—affects cost distribution and can alter marginal incentives. Free allocation can cushion firms, but output-based allocation may blunt the carbon price signal ([Burtraw et al., 2018](#)). Benchmark stringency is crucial: a tighter benchmark increases permit scarcity, while a looser one approaches a subsidy that offsets carbon costs ([Fischer, 2006](#)).

To analyze such policy design options and their interactions, the World Bank–IMF Climate Policy Assessment Tool (CPAT) provides a simplified but comprehensive framework. Initially

developed for carbon taxes and other mitigation policies, CPAT now handles ETS policies, allowing users to specify a cap (or intensity target) and allocation method and then computing the resulting carbon price, emissions path, and economic impacts. A key advantage is the ability to examine policy mixes—for example, an ETS in some sectors with a carbon tax in others—and how overlapping instruments affect permit demand and equilibrium prices.

CPAT also accounts for the uncertainty and price volatility of trading systems relative to fixed taxes. As discussed, the model applies a modest “volatility premium” to reflect that an ETS may need a slightly higher average price than a perfectly predictable tax to achieve the same emissions reduction.

This paper is structured as follows. Section 2 provides a typology of ETS and TPS instruments. Section 3 presents country case studies—EU, China, India, and the USA—with emphasis on power sector policies. Section 4 describes how CPAT models these instruments, including allocation regimes and integration with power sector modeling. Section 5 discusses policy implications. An Appendix offers technical details on the CPAT approach, including the equations behind ETS price–emissions calculations and the “engineer” power sector model for capacity investment, retirement, and dispatch.

2 Typology of ETS and TPS Instruments

2.1 Mass-based ETS vs. Intensity-based TPS

An ETS can be broadly categorized by the form of its emissions limit: **mass-based** (absolute) caps or **intensity-based** (relative) limits. A traditional cap-and-trade system sets a fixed quantity cap on emissions (e.g., tons CO₂ per year). Regulators issue a corresponding number of allowances (each typically 1 tCO₂) equal to the cap, and firms must surrender one allowance for each ton they emit. Because the total supply of allowances is fixed by the cap, the environmental outcome (total emissions) is predetermined, and the market finds a price such that aggregate abatement meets the cap. In theory, this price equilibrates the marginal cost of emission reduction across sources. If the cap is stringent, the allowance price will be higher, all else equal ([Burtraw et al., 2018](#)). A mass-based ETS thus provides certainty about the quantity of emissions but not the price, as the price will fluctuate with economic conditions, energy demand, and abatement costs.

By contrast, an **intensity-based** system (often called a **Tradable Performance Standard (TPS)** when applied to an industrial sector, or a rate-based ETS) does not fix total emissions, but rather sets a limit on emissions per unit of output (or per unit of activity). Regulated entities receive allowances in proportion to their output and a benchmark intensity. For example, under an intensity-based power sector ETS, a power plant might receive an allowance allocation equal to $\beta \times Q_i$, where β is a benchmark in tons CO₂ per MWh that declines over time. Each plant must surrender allowances equal to its actual emissions; firms that emit less than β per MWh will have surplus allowances to sell, while those exceeding the benchmark must buy extra. The market price of allowances adjusts such that the industry-wide average emissions intensity meets the target β ([Fischer, 2006](#); [ICAP, 2022a](#)).

However, unlike a fixed cap, total emissions in an intensity-based system can increase if output grows. The total allowance allocation in a compliance period is $\beta \times Q_{\text{total}}$; in aggregate this behaves like a *baseline-and-credit* system where the baseline is proportional to output. This design yields certainty about emissions intensity (emissions per unit output) but not about absolute emissions. It tends to stabilize compliance costs relative to output fluctuations: if

demand and output are higher than expected, more allowances are automatically issued (preventing price spikes, but leading to higher emissions), whereas in a recession output drops and fewer allowances are needed (preventing a price crash, but yielding lower emissions than expected). In essence, an intensity-based TPS shares features of price-based instruments by allowing quantities to vary, trading off some environmental certainty in exchange for potentially smoother prices and less impact on output.

Mathematically, under an intensity target, the compliance condition can be written for each firm (or unit) i as:

$$\text{Allowances allocated}_i = \beta \cdot Q_i,$$

$$\text{Allowances required}_i = E_i,$$

where Q_i is the output (e.g., MWh of electricity or tonnes of product) and E_i the CO₂ emissions of firm i . The firm must cover its emissions with allowances, so if $E_i > \beta Q_i$, it must purchase $E_i - \beta Q_i$ additional allowances; if $E_i < \beta Q_i$, it has $\beta Q_i - E_i$ surplus allowances to sell. In aggregate, total emissions $E_{\text{total}} = \sum_i E_i$ are not fixed, but the average emissions per unit output is constrained to β at equilibrium (assuming full compliance):

$$\frac{E_{\text{total}}}{Q_{\text{total}}} = \beta,$$

where $Q_{\text{total}} = \sum_i Q_i$. The market ensures that this intensity outcome is achieved by adjusting the permit price: a higher price induces more abatement (lower E_i relative to Q_i for each firm) but does not directly cap Q_i .

Because an intensity-based ETS expands the allowance supply in proportion to output, it acts similarly to a combination of a subsidy on output and a tax on emissions. For each unit of output, firms receive β allowances (the subsidy) but have to pay for their actual emissions. If a firm's actual emissions rate is $e = E_i/Q_i$, its net carbon cost per unit output is $P \times (e - \beta)$, where P is the allowance price (Fischer, 2006). Firms whose emissions rate is exactly at the benchmark ($e = \beta$) incur no net carbon cost on that output; firms cleaner than the benchmark ($e < \beta$) effectively receive a net credit (they can sell extra allowances), while dirtier firms ($e > \beta$) pay a penalty. At the margin, producing an extra unit of output increases emissions by

approximately e , but also earns β additional free allowances, so the net marginal cost of output is $P \times (e - \beta)$.

This means if the benchmark is generous (close to the actual emissions rate), the carbon cost per additional unit of output is very small, implying that the policy does little to deter higher output or demand growth. In the extreme case that the benchmark equals the status quo average e , the policy merely redistributes permits with no immediate incentive to reduce output or emissions intensity. Stricter benchmarks (smaller β) increase the portion of emissions that must be paid for, thereby increasing the effective carbon cost and inducing greater abatement. In Section 2.3 we discuss how output-based designs generally require higher carbon prices to achieve the same absolute emissions reduction as a fixed cap with full auctioning, since the output subsidy dilutes the price signal on production (Burtraw et al., 2018; Fischer, 2006).

It is worth noting that intensity-based trading systems are sometimes implemented under different names. A **Tradable Performance Standard (TPS)** typically refers to an intensity-based regulation in a specific sector. For example, a fuel economy standard with credit trading or an industrial energy efficiency benchmark with trading can be viewed as TPS policies. The Chinese national ETS for the power sector is effectively a TPS: each generator is assigned an emissions rate benchmark (tons CO₂ per MWh) based on unit type, and allowances are allocated proportional to actual generation at that benchmark rate (ICAP, 2022a). India's Perform, Achieve and Trade (PAT) scheme for industries is another example: it sets energy-intensity targets (energy use per unit output) for plants; those beating their target earn tradable Energy Saving Certificates, and those missing must purchase certificates (PMR, 2021). These intensity-based schemes are equivalent to output-based allocation in an ETS context (Fischer, 2006). In contrast, a classic cap-and-trade (like the EU ETS or RGGI in the US) is mass-based, with a fixed cap on total allowances issued.

2.2 Allowance Allocation: Auctioning vs. Free Allocation Designs

Orthogonal to the question of cap-based vs. intensity-based is the question of how allowances are distributed. **Auctioning** means firms must buy allowances (typically in government-run auctions), whereas **free allocation** gives some or all allowances to firms without charge, based on specified criteria. Free allocation can be done in several ways (Ellerman et al., 2010):

- *Grandfathering (historic baseline)* – Allowances are given in proportion to each firm’s past emissions or output. For example, a firm that emitted 100 tons in a base year might get 100 allowances (or some fraction thereof) each year. This rewards historically higher emitters and can raise equity concerns, and it necessitates rules for new entrants (who lack a history) ([Grubb et al., 2014](#)). Early phases of the EU ETS predominantly grandfathered allowances.
- *Benchmarking (fixed performance-based)* – Allowances are allocated according to a performance benchmark (e.g., industry best practice emissions rate) and an activity measure, but the activity is often an exogenous or historical level. For instance, a cement plant might get free allowances equal to an efficient benchmark (e.g., 0.8 tCO₂/ton cement) multiplied by its historical production capacity. More efficient (lower-emitting) installations receive no extra allowances beyond others of similar size, so unlike output-based allocation, benchmarking does not reward producing more output in the future; it simply rewards having lower emissions intensity relative to peers in the baseline period. Benchmarks are widely used for free allocation in later phases of the EU ETS, the California/Quebec system, and others ([ICAP, 2021](#)).
- *Output-Based Allocation (OBA)* – Allowances are distributed proportionally to current production output, multiplied by a benchmark or allocation rate per unit. This is a dynamic allocation: if a firm produces more, it gets more free allowances. It effectively acts as an output rebate or subsidy: every unit “earns” some allowances, offsetting some of the carbon cost of producing that unit ([Burtraw et al., 2009](#)). Output-based allocation is often motivated by competitiveness and leakage concerns for trade-exposed industries, as it helps maintain their production by lowering effective costs (they only pay for emissions above the benchmark per unit). It is used in some form in systems like California (for industrial sectors), New Zealand (for emissions-intensive trade-exposed sectors), and China (for the power sector) ([ICAP, 2022a](#)).
- *Intensity-based allocation* – This term is often used interchangeably with output-based allocation. Essentially, intensity-based free allocation means tying allowances to actual

output and an intensity standard (which is exactly how OBA is defined). For example, China’s power sector TPS gives each plant allowances = output $\times$ benchmark emissions intensity (ICAP, 2022a). We emphasize that in common usage, output-based allocation *is* intensity-based; both denote free allocation that scales with current production. (Some literature uses “intensity-based” to describe a policy where compliance is judged on an intensity metric rather than absolute emissions – which is exactly what a TPS is. In this paper we treat intensity-based allocation as synonymous with OBA as a mode of allocation.) (PMR, 2021)

In addition, hybrid approaches exist. **Partial auctioning** combined with partial free allocation is common. For instance, the EU ETS today auctions the majority of allowances (57% in 2021) but still gives a substantial share for free to industrial sectors based on benchmarks; the power sector in the EU generally must buy 100% of its allowances at auction (since Phase III), reflecting the policy choice that electricity producers can pass costs to consumers and should not receive windfall gains (Commission, 2021). Similarly, California’s cap-and-trade auctions about 60–65% of allowances, while granting the rest freely to industries (and also to electricity distributors, who must use the value for ratepayer benefit) (CARB, 2015). Most ETS have progressed toward greater auctioning over time, but free allocation remains prevalent globally. In fact, by emissions-weighted coverage, roughly 90–95% of emissions under ETS programs worldwide are still under regimes with significant free allocation, and only 5–10% under systems that fully auction allowances (ICAP, 2023b). The dominance of free allocation is largely due to the huge size of China’s ETS (which currently allocates nearly 100% of permits for free on an intensity basis) and other systems (like Korea’s) that also rely heavily on free benchmarking (ICAP, 2022b).

Impacts on incentives and prices: In theory, if allowances are freely given but fully tradable, firms should still face the same *opportunity cost* for emitting – every ton emitted is a permit they cannot sell – so the marginal incentive to abate is the same as under auctioning. Under this theory (the independence property), the cap determines the outcome, not the allocation method (Goulder et al., 2010). However, in practice, allocation affects behavior through several channels:

- Free allocation can create an “*endowment effect*” or liquidity effect: firms may treat freely received allowances as reducing their perceived cost of compliance, especially if they do not fully internalize the opportunity cost (Chan et al., 2012). Empirical studies have found that firms often do behave as if free permits lower their marginal cost, contrary to the pure theory (Martin et al., 2014).
- If allocation is output-based, there is an explicit incentive to increase output because doing so yields more free permits. This *dilutes the carbon price signal on the output decision*: a firm that expands production under OBA does not have to buy as many additional permits as it would under fixed allocation, since extra output comes with extra allowances (Burtraw et al., 2018). Thus, output-based allocation lowers the effective cost per unit output, tending to raise output compared to an equivalent scenario with auctioning or fixed free allocation. This effect can significantly reduce the emissions reduction achieved for a given carbon price, because higher output means higher emissions (even if intensity improves somewhat) (Fischer, 2006).
- With auctioning, by contrast, firms must pay for every ton from the outset, which generally leads to full cost pass-through into product prices (for sectors where demand allows) (Goulder et al., 2010). This can suppress demand for carbon-intensive goods and further reduce emissions, and it generates revenue that governments can recycle or invest (which itself can indirectly promote long-term emissions cuts, e.g., funding clean technology) (Parry, 2007; Vogt-Schilb et al., 2020). Free allocation yields no revenue and thus foregoes this opportunity.

Because of these differences, the same cap level can result in different carbon prices depending on allocation. If allowances are given free in an output-based manner, the market carbon price typically must rise higher to achieve a given emissions cut, compared to full auctioning, since firms face a weaker incentive per unit to cut back (Fischer, 2006; Goulder et al., 2010). In other words, free allocation (especially OBA) can be thought of as lowering the effective emissions elasticity of the system, requiring a higher price to reach the cap. Modeling studies confirm this: for example, one study found that under a fixed emissions cap, switching

from auctioning to output-based allocation in the EU cement sector nearly eliminated emission reductions – the carbon price would have to be extremely high to force the same cuts in output-based mode (Monjon and Quirion, 2011). In general, benchmark-based free allocation that is fixed (grandfathering or static benchmarking) does *not* dilute the marginal incentive as much as OBA, because once firms receive their fixed quota, producing an extra unit does not earn new allowances (Fischer, 2006). Thus, auctioning and fixed benchmark allocation yield similar price signals for incremental abatement, whereas fully dynamic OBA significantly blunts the signal on the extensive margin (production/output) (Goulder et al., 2010).

TABLE 1. Indicative carbon price required to deliver a given emissions cut (e.g., 10%) under alternative allocation regimes. Output-based allocation notably dampens the price signal, requiring a higher permit price for the same abatement.

Allocation Type	Price for 10% Cut (vs. auction)	Effect on Marginal Incentives
Auctioning or Fixed Benchmarking	$\approx 1.0\times$	Full opportunity cost; strong incentive on all tonnes.
Grandfathering (historic free)	$\approx 1.0\times$ (theoretical); slightly lower in practice	Opportunity cost preserved but may weaken investment signals; possible windfall issues.
Output- or Intensity-Based	$\approx 1.5\text{--}2.0\times$ (or higher)	Functions as output subsidy; only emissions above benchmark face full cost.

Beyond the static considerations of who pays and how much is abated, allocation has **dynamic and distributional implications**. Auctioning, by raising revenue, creates opportunities to use that revenue productively – for example, funding clean innovation or compensating households. Free allocation foregoes that revenue, instead benefiting the owners of regulated firms (at least in the short run). This can affect political economy: free allocation is often used to gain industry support or avoid international leakage of emissions-intensive production (Aldy, 2011). However, free allocations, especially if output-based, may slow down long-term decarbonization by reducing firms’ incentives to invest in lower-carbon processes. Empirical evidence from the EU ETS shows that in phases with more auctioning, firms directed more resources toward efficiency and low-carbon innovation, whereas in the early phases with mostly free allocation, there was less low-carbon investment beyond what was required to meet the cap (Martin et al., 2014). The output-subsidy effect of OBA tends to favor incumbent high-emitting producers by shielding them from costs, which can dampen the market for cleaner

alternatives ([Fischer, 2006](#)). Thus, while free allocation can be a useful transition measure to address competitiveness concerns, the trend is generally toward phasing it down and increasing auction shares over time in many systems ([ICAP, 2023b](#)).

In summary, ETS designs span a spectrum from pure quantity control with full auctioning (delivering a clear price signal and revenue, but potentially higher costs for industry) to intensity-based, output-linked systems that soften the impact on industry at the expense of higher carbon prices (for a given target) and less certainty on emissions. The CPAT tool is built to handle both types of systems. In the next section, we explore how these design choices manifest in four jurisdictions, before turning to how CPAT models them.

3 Power and Industry Designs

To illustrate the above concepts, we examine four major jurisdictions—the European Union, China, India, and the United States—focusing on their power sector carbon pricing instruments and any parallel industrial emissions trading or performance standard policies. These cases highlight the diversity of approaches: the EU’s cap-and-trade with increasing auctioning, China’s intensity-based power sector ETS, India’s energy intensity trading scheme for industries, and the mix of regional programs in the US. Table 2 at the end of this section provides a comparative summary of ETS/TPS design features in these jurisdictions’ power and industry sectors.

3.1 European Union

The EU ETS, launched in 2005, is the world’s largest cap-and-trade system by coverage outside of China. It covers power generation and large industrial emitters across EU member states (and is linked with Norway, Iceland, and Liechtenstein). The EU ETS is a classic mass-based system: an EU-wide cap (currently about 1.3 billion tCO₂ in 2025, declining annually) sets the total allowances. **Allocation:** In its initial Phase I (2005–07) and Phase II (2008–12), the EU ETS relied almost entirely on free allocation via grandfathering based on historical emissions (Ellerman et al., 2010). This led to over-allocation and windfall profits for some utilities. Starting in Phase III (2013–2020), the EU moved decisively toward auctioning: electric power generators now receive almost no free allowances and must buy virtually 100% of what they need (Commission, 2021). Industrial sectors still receive a significant free allocation to address competitiveness concerns, but it is now based on benchmarks rather than grandfathering. These benchmarks are defined in terms of emissions per unit output (e.g., tCO₂/ton of steel), but the allocation for a given plant is essentially fixed *ex ante*. In Phase IV (2021–2030), the free allocation for industry is being gradually reduced for sectors that are deemed less at risk of carbon leakage, and more allowances are auctioned. Currently, about 57% of total EU ETS allowances are auctioned; the power sector is nearly 100% auctioned, and industry is majority free-allocated.

Impacts and interactions: The EU’s choice to auction power sector allowances means the

carbon price is fully passed through into electricity prices, contributing to shifting generation from coal to gas and zero-carbon sources over time. The presence of parallel policies—notably renewable energy targets and efficiency mandates—interacted with the ETS. A surplus of allowances and low prices during earlier phases prompted creation of the Market Stability Reserve. Revenues from auctioning have been significant and used for innovation and state-level programs(Commission, 2020). Beginning in 2021, industrial allocations are adjusted if output varies more than 15% to introduce partial output responsiveness. The EU also plans to phase in a carbon border adjustment mechanism (CBAM) after 2026.

In summary, the EU ETS exemplifies a cap-and-trade system with a gradually tightening cap and a shift from free allocation to auctioning. The power sector policy closely resembles a carbon tax in incidence, while the industrial policy balances cost and competitiveness via benchmarking(Commission, 2018). The EU experience shows the importance of a binding cap and use of revenues to support innovation.

3.2 China

China launched its national ETS in 2021, initially covering the power sector (over 2,000 plants, 4 GtCO₂). China's system is an **intensity-based tradable performance standard**. Regulators set CO₂/MWh benchmarks by generator type. Each plant receives free allowances = benchmark emissions rate × electricity generation output(Zhang et al., 2022). For example, a coal plant benchmarked at 0.85 tCO₂/MWh producing 1,000 MWh gets 850 allowances. Surpluses can be sold; deficits require purchases. There is no auctioning: 100% of allowances are freely allocated(of Ecology and Environment, 2021). The benchmarks are expected to tighten over time.

Behavior and price: Because allocation scales with output, the system has little impact on electricity prices or dispatch decisions. It rewards efficiency but does not constrain total generation. The carbon price started low (\$7/ton) due to weak stringency. As benchmarks tighten, more plants face allowance shortfalls. Banking is allowed for flexibility.

This design permits emissions to grow with output, challenging absolute peaking goals. The ETS interacts with renewable mandates and coal controls. The system may evolve toward including an absolute cap. Expansion to sectors like steel and cement is expected, using sectoral

intensity benchmarks([ICAP, 2023a](#)).

Modeling China's ETS involves applying a price that yields a desired emissions intensity, with output determining total emissions. Because prices are not passed through to consumers (due to free allocation and regulated tariffs), supply-side efficiency improvements dominate responses.

3.3 India

India lacks a CO₂ cap-and-trade but implemented an **intensity-based market mechanism** known as the **Perform, Achieve, and Trade (PAT)** scheme. Launched in 2012, it sets energy efficiency targets for large emitters across sectors. Entities outperforming targets earn Energy Saving Certificates (ESCs); underperformers must purchase them. This baseline-and-credit system improves energy intensity.

Although PAT is not a CO₂ cap, it contributes to reductions by curbing energy use. Trading volumes and ESC prices have been modest. India is now developing a **Carbon Credit Trading Scheme (CCTS)** to cover nine sectors, assigning tCO₂/unit targets over 6-year periods([Government of India, 2023](#)). Surpluses yield tradable Carbon Credit Certificates (CCCs), and shortfalls require purchasing CCCs. The system includes Scope 2 emissions, so cleaner power generation benefits industries.

India's approach aligns with its national 2030 goal: a 45% GDP emissions intensity reduction (vs. 2005). The scheme allows growth while promoting efficiency. Modeling it resembles China's TPS: carbon pricing is on intensity deviations. PAT had little product price impact. While the power sector is not directly under CCTS, its emissions factor matters for industrial compliance.

3.4 United States

The US has no federal ETS but features regional programs:

- **RGGI** (Regional Greenhouse Gas Initiative): 12 Northeast/Mid-Atlantic states, power sector only. Mass-based cap (96 MtCO₂) declining 3%/yr. Over 90% of allowances auctioned([RGGI, Inc., 2021](#)). Proceeds support efficiency and renewables. Price floors and reserves stabilize prices. Recent prices range \$3–\$13/ton.

- **California Cap-and-Trade:** Economy-wide since 2013 (linked with Quebec). Covers 75–80% of state GHGs. Absolute cap declining to meet -40% by 2030. Around 60–65% of allowances are auctioned; industrials get free allocation via output-based benchmarks(CARB, 2015). Utilities receive free allowances for ratepayer protection. The price floor (\$20) and containment reserves stabilize prices (currently \$30/ton).

Other developments include Washington State’s 2023 program. Federal TPS examples include the Renewable Fuel Standard and the former Clean Power Plan’s proposed rate-based trading.

Power sector focus: US decarbonization stems from cap-and-trade, regulations, and market shifts. Auction revenue in RGGI and California funds additional mitigation. Cap-and-trade in power can be modeled similarly to a carbon tax; overlapping policies influence cap outcomes (“waterbed” effect). Surging demand tightens caps and raises prices unless adjusted.

These case studies underscore how policy objectives shape ETS design. The EU prioritizes carbon pricing and revenue; China phases in stringency while protecting growth; India focuses on efficiency; and the US demonstrates diversity in subnational implementation. Modeling must reflect cap type, price pass-through, and policy overlap.

In the next section, we examine how the Climate Policy Assessment Tool (CPAT) incorporates these design features, especially in its power sector model.

TABLE 2. Comparison of ETS or TPS design features in select jurisdictions. EU and US use absolute caps in power; China and India use intensity-based designs. EU uses fixed benchmarking; California uses output-based allocation. China and India have 100% free allocation; EU and US auction significant shares.

Country	Power Sector ETS/TPS Design	Industrial Sector ETS/TPS Design
European Union	EU ETS Phase IV: absolute cap. <i>Allocation:</i> 100% auctioning for powerPower emissions under carbon price; revenues fund innovation.	Covered under same cap. <i>Allocation:</i> Free allocation via fixed product benchmarks(Commission, 2011). 40–45% of industrial allowances free; remainder auctioned. Partial output-based adjustments for large output changes.
China	National ETS (since 2021) for power. <i>Cap:</i> Intensity-based (rate): benchmark $\times$ actual output(of Ecology and Environment, 2021). <i>Allocation:</i> 100% free, output-proportional(Zhang et al., 2022). Price low; minimal effect on retail electricity prices.	Expansion to industry planned: sector-specific intensity benchmarks. Past pilots had mixed caps. Free allocation to continue initially (no auctions yet).
India	No direct cap-and-trade. Power sector in PAT for efficiency targets; CCTS includes Scope 2.	PAT evolving to CCTS covering nine sectors. <i>Cap:</i> Intensity-based(Government of India, 2023). <i>Allocation:</i> Free credits for over-performance; no auctions. Targets are output-relative; trading of certificates among firms.
United States	No federal ETS. RGGI: power only, cap declining 3%/yr; 90% auctioned(RGGI, Inc., 2021). California: power under broader cap; partial auctioning; price stability provisions.	California covers industry under cap. <i>Allocation:</i> Output-based free allocation for leakage-prone sectors(CARB, 2015). Other states follow similar design; future federal policy uncertain.

4 CPAT Modeling of ETS Policies

The CPAT tool’s mitigation module is designed to analyze carbon pricing policies in a simplified, data-driven manner across countries. Initially focused on carbon taxes, the module has been extended to handle ETS by introducing a mechanism to solve for the equilibrium permit price given an emissions target (cap or intensity goal) and to account for allocation differences, policy interactions, and price volatility. This section provides an overview of how CPAT models ETS, including both the general algorithm and specific considerations for the power sector.

4.1 Solving for Permit Price under a Cap or Target

In a carbon tax scenario, the price is exogenous and emissions are computed by the model. In an ETS, the emissions outcome is set by the cap (or intensity target), and the price is endogenous. CPAT approaches this with a two-step solution:

1. A **pre-model approximation** quickly estimates the permit price needed to meet the cap, using reduced-form elasticities. This gives an initial guess for the carbon price.
2. A more detailed **iterative solver** then “goal-seeks” the precise permit price by running the full sectoral simulation (including the power model, etc.) to match the cap. This is essentially a numerical root-finding to ensure emissions equal the cap, and it is slower (on the order of minutes) but more accurate.

The pre-model in CPAT assumes a log-linear response of emissions to carbon price. For each sector i , a “half-elasticity” n_i is defined such that a small carbon price (say \$1 per ton) reduces emissions by n_i percent. This n_i is calibrated from empirical and engineering data for each sector. If the user sets an ETS cap that requires a certain percentage reduction $X\%$ below baseline emissions in the covered sectors, CPAT can invert the elasticity relation to estimate the price P that would achieve that reduction:

$$\ln\left(\frac{E_{\text{with ETS}}}{E_{\text{baseline}}}\right) = -n_{\text{agg}} \cdot P,$$

where n_{agg} is an aggregate elasticity for the sectors covered by the ETS . Solving gives

$$P \approx \frac{1}{n_{agg}} \ln\left(\frac{1}{1-X}\right),$$

for small X . This is a simplification, but it provides a starting price. The pre-model actually runs twice: once to account for other policies' effects (see next paragraph) and second to solve for the ETS price.

Policy interactions: If there are other policies alongside the ETS (e.g., a carbon tax on other sectors, a renewable energy policy, fuel taxes, etc.), CPAT first computes the emissions reduction from those policies alone, then subtracts that from the required total reduction, leaving the residual that the ETS must achieve. For example, if the overall target is a 20% reduction and an existing carbon tax covering half the emissions already achieves 5% reduction, the ETS cap would need to cover the remaining 15% reduction. The pre-model would then estimate the ETS price for that residual. This ensures consistency when multiple instruments overlap. It effectively assumes other policies are enforced first, reducing baseline emissions, and the ETS then caps the rest.

Once the pre-model provides an initial permit price, the full model is run. This entails computing energy demand and supply in each sector, running the detailed power sector model (described below), etc., with that carbon price applied to fuels/emissions in covered sectors. The resulting emissions are aggregated and compared to the cap:

- If emissions exceed the cap, the price guess was too low; it is increased.
- If emissions are below the cap (too much abatement), the price was higher than needed; it is reduced.

CPAT iterates until the emissions outcome matches the cap (within tolerance). This goal-seeking converges in a few iterations since the emissions vs. price relationship is monotonic.

For intensity-based targets (TPS), the procedure is slightly different: rather than matching an absolute emissions cap, the model aims to match an average emissions per output (e.g., power sector CO_2/MWh). In principle, this also can be solved by iteration: adjust the price until the model's output intensity equals the target. Under the hood, one can translate an in-

tensity target into an implied absolute emissions for the projected output level. CPAT’s current implementation primarily handles absolute caps, but an intensity target for power or industry can be modeled by converting it to an equivalent cap for the given output. The nuance is that if the carbon price changes output significantly, the intensity target might need recalculation. However, in systems like China’s where output effect is small (due to OBA shielding the consumer price), one can treat output as fixed for initial estimation. More generally, CPAT could iterate on intensity similarly to find the equilibrium price where E/Q meets the standard.

Volatility and “ETS price premium”: A novel aspect in CPAT’s approach is recognizing that an ETS’s price may need to be higher than an equivalent carbon tax to achieve the same outcome, due to price volatility and associated inefficiencies (Weitzman, 1974). As discussed earlier, uncertain permit prices can lead firms to delay investment and result in suboptimal timing of abatement. CPAT introduces a parameter ϕ to represent the fractional increase in required price for an ETS relative to a perfectly predictable price. Based on literature, CPAT uses a default $\phi \approx 10\%$, meaning an ETS needs roughly a 10% higher price to achieve the same emissions cut as a stable carbon tax. In practice, CPAT applies a “volatility factor” of about 0.9 to the effective elasticity for ETS; essentially, for a given price, the model assumes an ETS achieves only 90% of the emissions reduction that a tax would, requiring a higher price to hit the cap. This aligns with studies showing a 10–15% cost premium for quantity instruments (Pizer, 2002). The user can adjust this factor if, for example, strong price control measures (like an ETS price floor) are in place or if banking and borrowing are allowed, which tend to reduce volatility impacts (Fell and Morgenstern, 2010). By default, CPAT’s inclusion of this factor means that for the same cap, the tool might project a slightly higher permit price than a naive model would, reflecting economic theory and empirical findings on volatility’s cost impact (Aldy, 2011).

4.2 Representing Allocation: Auctioning vs. Output-Based in CPAT

While CPAT does not explicitly simulate each allocation type’s micro-level dynamics, it provides a simplified switch to capture the two extremes of allocation impact:

- **Auctioning (or fixed free allocation):** In this mode, the model assumes the carbon price is fully internalized by firms and passed on in product prices. This affects demand (ex-

tensive margin) as well as emissions intensity choices (intensive margin). Also, auction revenue is calculated (carbon price $\times$ emissions) which can feed into fiscal results, though the use of revenue is external to the core emissions model.

- **Free Allocation with output-basis (“non-auctioning” in CPAT):** In this mode, the model assumes the policy functions like an output-based rebate, meaning firms’ marginal cost of production is reduced relative to the auction case. In CPAT’s structure, this is implemented by effectively zeroing out the carbon price impact on consumer prices (no extensive margin effect) and reducing government revenue accordingly (since any payments are rebated). In other words, CPAT treats output-based or generous free allocation as a scenario where the carbon price only incentivizes reducing emissions intensity (fuel switching, efficiency) but does not curtail output, and it generates no net revenue (like a feebate system). This simplification approximates the maximal dampening of the price signal that free allocation can cause. Grandfathering and benchmarking, in reality, lie somewhere between full auction and full OBA in terms of incentive effects; however, to keep the tool user-friendly, CPAT offers essentially two options: assume full price pass-through (auctioning) or assume no price pass-through (output-based rebate).

This binary approach is admittedly a simplification, but it reflects the observation that many ETS either have significant output-based components or they don’t. CPAT’s documentation notes that the allocation approach could be refined in future to allow intermediate cases (e.g., partial auctioning, or fixed free allocation that still yields full incentive but no revenue). For now, if a user wants to analyze a mixed system, they might run two scenarios to see the range: one as if fully auctioned, one as if fully output-rebated, to understand the bracketed outcomes.

To illustrate the impact: if a country chooses output-based allocation for industry, CPAT’s non-auction mode will show a higher permit price needed to achieve the cap, compared to auction mode. It will also show zero revenue raised from the ETS (since free allowances generate no government revenue). Emissions reduction will be achieved more via intensity improvement than output reduction. By contrast, in auction mode, part of the emissions reduction may come from demand/output decreasing (e.g., electricity price rises causing lower consumption), and substantial revenue is collected. The tool thus helps quantify the trade-off: output-based

allocation can shield output (avoiding, say, a dip in GDP or production in the short run) but it means needing a steeper carbon price to hit the same emissions outcome .

5 Policy Implications and Conclusions

Our exploration of ETS and TPS designs and their modeling in CPAT leads to several insights relevant to policymakers and modelers:

1. Instrument choice (cap vs intensity) involves trade-offs: A mass-based cap-and-trade guarantees environmental outcome (emissions quantity) but results in uncertain and potentially volatile prices. An intensity-based system provides more certainty that industries will meet an efficiency goal and tends to yield more stable prices relative to economic output swings, but it cannot guarantee a particular emissions level and tends to require higher prices (or additional policies) to achieve the same absolute reduction. Policymakers need to consider whether their priority is a fixed emissions trajectory or managing economic impacts. For developing countries concerned about allowing growth, intensity targets offer flexibility – as seen in China and India – but if the ultimate objective is to reduce absolute emissions, the intensity approach may need to evolve into a cap as growth slows. CPAT can help quantify how much more emissions might occur under an intensity target versus an absolute cap in a high-growth scenario.

2. Allocation design is not just distributional, it's environmental: While free allocation is often viewed as a way to address competitiveness and equity (who pays), it also affects emissions and prices. Output-based allocation, in particular, acts as a production subsidy that can lead to higher output and emissions than otherwise, meaning the cap has to be tighter (or the price higher) to compensate ([Goulder et al., 2010](#); [Burtraw et al., 2002](#)). If the policy goal is to hit a certain emissions reduction at least cost, theory favors auctioning. However, if the goal is to reduce leakage or protect jobs in trade-exposed sectors, output-based allocation may be justified, accepting that it will dilute emissions impacts in the short run. The case studies show different approaches: the EU moved to auctioning for power, enhancing the carbon signal, whereas California retained output-based free allocation for industry to prevent leakage. One implication is that jurisdictions with output-based allocation might consider complementary measures (higher carbon price floors, additional sector-specific standards) since the pure price signal is weaker. CPAT results make this clear, as switching the allocation mode in the model significantly changes the required carbon price and the distribution of burden (government revenue vs. industry cost).

3. Policy mixes and interactions matter: Emissions trading does not occur in a policy vacuum. Renewable energy mandates, efficiency standards, fuel taxes, and other measures can either complement or interfere with an ETS. A key example is the “waterbed effect” in cap-and-trade: if a parallel policy (say a renewable subsidy) lowers emissions in a capped sector, the cap means total emissions remain fixed – the extra reduction simply translates to a lower permit price and possibly less effort elsewhere. This can be mitigated by mechanisms like the EU’s Market Stability Reserve which can adjust the cap in response to sustained demand drop for allowances. In intensity systems, overlaps are a bit different: extra renewables might overachieve the intensity target and make the permit price zero (because the target is met), effectively causing the TPS to cease binding. Thus, policy interaction can lead to redundant policies unless instruments are designed to reinforce each other. CPAT’s ability to include multiple policies allows users to test interactions: for instance, adding a clean power standard on top of an ETS and observing the change in equilibrium carbon price and emissions. The model underlines that to maximize effectiveness, overlapping policies should ideally address market failures not covered by the carbon price (like innovation spillovers) or aim to raise ambition (tighten the cap when adding complementary policies).

4. Power sector transformation is critical and benefits from explicit modeling: The engineer model in CPAT, though simplified, highlights that short-run fuel switching and long-run capacity decisions both contribute to mitigation. An ETS can incentivize immediate gains (like running gas instead of coal, if gas is available) and longer-term investments (like renewables, storage). But these depend on the presence of alternatives and the elasticity of substitution in the power system. For example, in some developing countries, gas capacity may be limited, so a carbon price might not reduce coal generation much until renewables scale up – making near-term price effects small and long-term investment effects more important. The model can show if an ETS alone would cause significant changes or if complementary measures (like renewable energy investments funded by auction revenue) are needed to actually shift the generation mix. The case studies support this: the EU’s carbon price has worked in tandem with renewables policy to squeeze out coal; China’s intensity-based ETS alone might not push coal out rapidly without renewables targets and coal caps also in place.

5. Importance of gradual transitions and flexibility mechanisms: Real-world ETS design often incorporates free allocation initially and transitions to auctioning, as well as cost containment measures (price floors, ceilings, banking). These features reflect political and economic realities – industries need time to adjust, and extremely volatile prices are undesirable. Our analysis indicates that having a predictable rising price (through a tax or a well-regulated ETS) tends to promote investment better than wild swings (Aldy, 2011). Systems like California’s show that a price floor can ensure a minimum investment signal, and revenue recycling can build support. Meanwhile, free allocation can be phased down as border carbon adjustments or international agreements level the playing field. For new ETS adopters (like some developing countries), starting with a modest intensity-based system or a pilot with free allocation might make sense to build MRV capacity and industry acceptance; over time, they can tighten toward an absolute cap and increase auctioning, following the trajectory of the EU or RGGI. CPAT can be used to map out such pathways, showing, for instance, how the carbon price might evolve as free shares are reduced and how emissions outcomes improve.

In conclusion, emissions trading systems are versatile tools that can be tailored to different policy objectives – whether prioritizing environmental certainty, economic stability, or competitiveness. The Climate Policy Assessment Tool provides a means to quantitatively assess these design choices and their interactions in a consistent modeling framework. By comparing scenarios (cap vs intensity, auction vs free, etc.), policymakers can better understand the implications before implementation. As more countries consider carbon markets (often in conjunction with other measures under the Paris Agreement pledges), such analysis is invaluable to design effective and coherent climate policy packages. The case studies of the EU, China, India, and the US demonstrate that there is no one-size-fits-all: each has adopted a mix that suits its circumstances. Nonetheless, the fundamentals of carbon pricing economics apply universally, and tools like CPAT help illuminate them: ultimately, who pays the carbon cost, and how that cost incentivizes changes in energy systems, will determine how successful we are in reducing emissions. Robust modeling and careful design can ensure that ETS and TPS instruments play a strong role in the global effort to mitigate climate change.

Appendix: Power Sector Modeling Integration

The power sector is often the linchpin of mitigation analysis, as it is usually the largest source of emissions and the one with the widest array of abatement options (fuel switch, renewables, demand response, etc.). CPAT recognizes this by incorporating two different models for power supply behavior: (1) an **“engineer” model**, which simulates capacity planning, dispatch, and retirements in a simplified way, and (2) an **elasticity-based model**, which uses aggregate substitution elasticities to approximate changes in the generation mix. The user can choose which model to use; the default is often an average of both, to reflect uncertainty. Here we focus on the engineer model, as it provides more structural detail, especially relevant for understanding policy interactions in the power sector (Appendix B provides the full methodological description).

In CPAT’s engineer power model, the key mechanism is that a carbon price (from an ETS or carbon tax) affects the variable cost of fossil fuel generation, which in turn influences dispatch (which plants generate) and investment (which new capacity is built). The model takes as inputs:

- Projected electricity demand growth (derived from GDP growth, price elasticity of demand, etc.).
- Existing capacity by technology (coal, gas, oil, nuclear, various renewables) and assumptions on retirements.
- Technical and economic parameters (efficiency, capacity factors, fuel costs, capital costs of new builds, etc.).

The engineer model then simulates year-by-year:

1. **Demand determination:** Based on GDP and electricity price trends, demand for electricity is forecast. A carbon price can raise generation cost and thus price, feeding back to slightly reduce demand growth (though in an ETS with OBA, this feedback might be muted because price pass-through is less).

2. **Existing capacity and retirement:** Each year, some existing plants retire. CPAT assumes a lifetime for each type (e.g., coal 40 years unless policy accelerates, renewables 25 years). Non-coal capacity is often assumed to retire at a steady rate (1/lifetime per year). Coal may have explicit retirement schedules or an economic test for early retirement if carbon costs make it unprofitable.
3. **Capacity expansion (investment):** If projected demand (plus reserve margin) exceeds available generation, new capacity must be built. CPAT determines new capacity by looking at the gap between expected demand and what existing plus committed plants can supply, accounting for capacity factors. The model computes total generation requirement and thus capacity needed. It then allocates this new capacity across technologies based on relative cost-competitiveness. Specifically, CPAT uses a logit choice model: it calculates the levelized cost of electricity (LCOE) for each option (including capital, fuel, O&M, and carbon costs) and then assigns market share of new builds probabilistically:

$$s_i = \frac{\exp(-\kappa LCOE_i)}{\sum_j \exp(-\kappa LCOE_j)},$$

where κ is a calibration parameter. This reflects imperfect cost minimization due to constraints or preferences. Carbon pricing raises the LCOE of fossil fuels, favoring low-carbon capacity.

4. **Dispatch (utilization):** After capacity is determined, the model allocates generation among available plants to meet demand. Dispatchable sources (coal, gas, oil) are dispatched based on short-run marginal cost (fuel + carbon cost). CPAT uses another logit formulation for the split between coal and gas generation as a function of their relative operating costs. Non-dispatchable generation (wind, solar) is assumed to produce at maximum availability (capacity factor). Nuclear and must-run hydro are prioritized. The dispatch logit might be:

$$\frac{G_{\text{coal}}}{G_{\text{gas}}} = \exp(-\lambda \Delta MC),$$

where ΔMC is the difference in marginal cost (including \$ per MWh carbon cost) between

coal and gas, and λ is a sensitivity parameter.

5. **Emissions calculation:** Given generation by each fuel and emissions factors, the model computes total CO₂ emissions from power. This feeds back into the overall ETS accounting. If the power sector is under an ETS cap in CPAT, the iterative solver adjusts the carbon price until emissions meet the cap.

The engineer model thus captures both short-term substitution (dispatch) and long-term investment shifts due to carbon pricing . It distinguishes dispatchable from non-dispatchable sources, which the simpler elasticity model does not. However, it operates at annual resolution with simplified load duration, using typical capacity factors rather than full chronological dispatch .

CPAT's ETS-power integration enables policy interaction analysis. For example, one can add a renewable portfolio standard (RPS) by exogenously increasing renewable capacity and observe the effect on permit prices. Typically, an RPS lowers emissions and permit demand, thus reducing the ETS price unless the cap is tightened. Such interactions can be tested via scenarios. CPAT's Output Panel shows how ETS carbon price, power mix, emissions, and other indicators vary under different allocation modes and policy packages.

Finally, CPAT's power model reflects differences in allocation regimes via electricity prices. Under auctioning, the full carbon price enters generation costs and retail prices (fuel cost + $P \times$ emissions factor). Under output-based allocation, CPAT assumes a feebate structure: carbon costs influence dispatch (firms still minimize emissions per MWh), but a per-MWh subsidy offsets consumer price increases. Hence, retail prices rise less under non-auctioning. This mirrors practice (e.g., California), where carbon costs are passed through under auctioning but may be muted under output-based schemes.

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